Homework 5

EE 999 — Introduction to Control Systems

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Due Wednesday, October 14, 2026

Problem 1.A — Transfer function

$$U(s) = k_p R(s) \tag{1}$$
$$Y(s) = \frac{b_1}{s + a_1} U(s) \tag{2}$$
$$Y(s) = \frac{k_p b_1}{s + a_1} R(s) \tag{3}$$

$$G_{ry1}(s) = \frac{Y(s)}{R(s)} = \frac{k_p b_1}{s + a_1} \tag{4}$$

Problem 1.B — Transfer function

$$Y(s) = \frac{b_1}{s + a_1} U(s) \tag{5}$$
$$U(s) = \frac{k_p s + k_i}{s} R(s) \tag{6}$$
$$Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} R(s) \tag{7}$$

$$G_{ry2}(s) = \frac{Y(s)}{R(s)} = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} \tag{8}$$

Problem 1.C.1 — Transfer function

$$Y(s) = \frac{b_1}{s + a_1} U(s) \tag{9}$$
$$U(s) = \frac{k_p s + k_i}{s} E(s) \tag{10}$$
$$E(s) = R(s) - Y(s) \tag{11}$$
$$Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} E(s) \tag{12}$$

$$Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} (R(s) - Y(s)) \tag{13}$$
$$\left(1 + \frac{b_1(k_p s + k_i)}{s^2 + a_1 s}\right) Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} R(s) \tag{14}$$
$$\frac{s^2 + (a_1 + b_1 k_p)s + b_1 k_i}{s^2 + a_1 s} Y(s) = \frac{b_1(k_p s + k_i)}{s^2 + a_1 s} R(s) \tag{15}$$
$$Y(s) = \frac{b_1 k_p s + b_1 k_i}{s^2 + (a_1 + b_1 k_p)s + b_1 k_i} R(s) \tag{16}$$

$$T_{ry3}(s) = \frac{Y(s)}{R(s)} = \frac{b_1 k_p s + b_1 k_i}{s^2 + (a_1 + b_1 k_p)s + b_1 k_i} \tag{17}$$

Problem 1.C.2 — Assign Values

$b_1 = 3,\ a_1 = 2,\ k_p = 1,\ k_i = 5$

$$T_{ry3}(s) = \frac{(3)(1)s + (3)(5)}{s^2 + ((2) + (3)(1))s + (3)(5)} \tag{18}$$

$$T_{ry3}(s) = \frac{3s + 15}{s^2 + 5s + 15} \tag{19}$$

Problem 1.C.3 — Pole and zero locations

$$p_{1,2} = -2.5 \pm 2.958j \tag{20}$$

$$z_1 = -5 \tag{21}$$

Problem 1.C.4 — Step response

Code: `hw_05_1_C_4.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

b_1 = 3;
a_1 = 2;
k_p = 1;
k_i = 5;

num = [b_1*k_p, b_1*k_i];
den = [1, (a_1 + b_1*k_p), b_1 * k_i];

T_ry = tf(num, den);

T_ry_poles = pole(T_ry)
T_ry_zeros = zero(T_ry)

figure;
step(T_ry);
grid on;
title("Step response of T_{Ry3}(s)");

[yy, tt] = step(T_ry);

T_ry_overshoot = max(yy) - 1
```

Output: `output/hw_05_1_C_4.txt`

```
T_ry_poles =

  -2.5000 + 2.9580i
  -2.5000 - 2.9580i

T_ry_zeros = -5
T_ry_overshoot = 0.1134
```

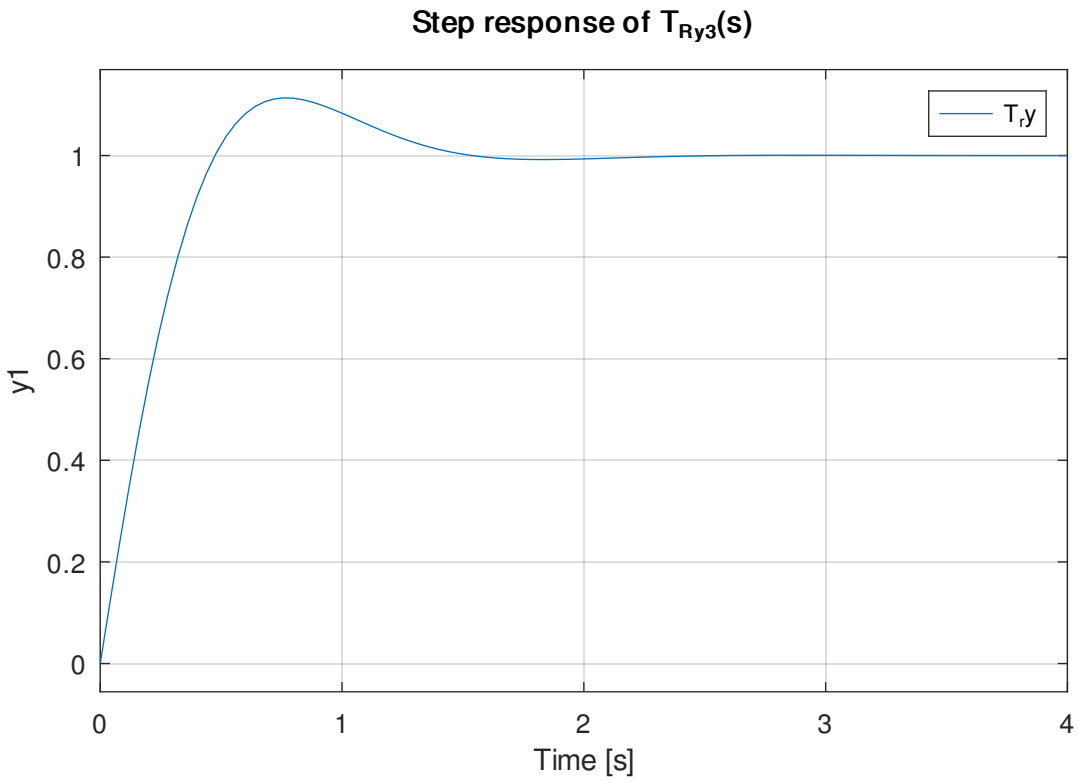


Figure 1: Step response of $T_{ry3}(s)$ with $k_p = 1$, $k_i = 5$.

Problem 1.C.5 — Overshoot

from above code, overshoot is 11.34%

Problem 1.C.6 — Adjusted k_p

Code: hw_05_1_C_6.m

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

b_1 = 3;
a_1 = 2;
k_p = 1.6;
k_i = 5;

num = [b_1*k_p, b_1*k_i];
den = [1, (a_1 + b_1*k_p), b_1 * k_i];

T_ry = tf(num, den);

T_ry_poles = pole(T_ry)
T_ry_zeros = zero(T_ry)

figure;
step(T_ry);
grid on;
title("Step response of T_{Ry3}(s)");

[yy, tt] = step(T_ry);

T_ry_overshoot = max(yy) - 1
```

Output: output/hw_05_1_C_6.txt

```
T_ry_poles =

  -3.4000 + 1.8547i
  -3.4000 - 1.8547i

T_ry_zeros = -3.1250
T_ry_overshoot = 0.044085
```

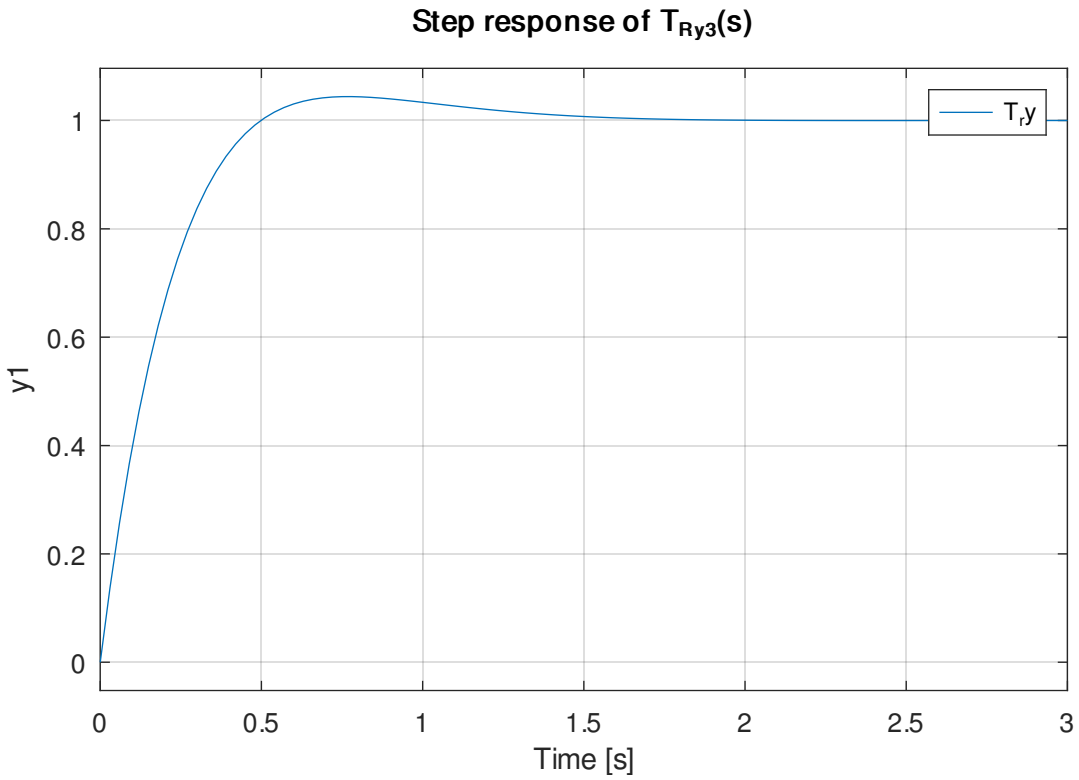


Figure 2: Step response of $T_{ry3}(s)$ with $k_p = 1.6$, $k_i = 5$.

Yes the overshoot is less than 5% at 4.41%.

Problem 2 — Pole and zero locations

Code: `hw_05_2.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

num_1 = [1 2];
den_1 = [1 10 8];
G_1 = tf(num_1, den_1);
G_1_poles = pole(G_1)
G_1_zeros = zero(G_1)
disp("-------------------");

num_2 = [5 15];
den_2 = [1 2 10];
G_2 = tf(num_2, den_2);
G_2_poles = pole(G_2)
G_2_zeros = zero(G_2)
disp("-------------------");

num_3 = [1 3];
den_3 = [1 -4 8];
G_3 = tf(num_3, den_3);
G_3_poles = pole(G_3)
G_3_zeros = zero(G_3)
disp("-------------------");

num_4 = [5 15];
den_4 = [1 2 6 10];
G_4 = tf(num_4, den_4);
G_4_poles = pole(G_4)
G_4_zeros = zero(G_4)
disp("-------------------");

num_5 = [2 9];
den_5 = [1 2 6 30];
G_5 = tf(num_5, den_5);
G_5_poles = pole(G_5)
G_5_zeros = zero(G_5)
disp("-------------------");
```

Output: `output/hw_05_2.txt`

```
G_1_poles =

  -9.1231
  -0.8769

G_1_zeros = -2
-------------------
G_2_poles =

  -1 + 3i
  -1 - 3i

G_2_zeros = -3
-------------------
G_3_poles =

   2 + 2i
   2 - 2i

G_3_zeros = -3
-------------------
G_4_poles =

  -0.1090 + 2.3664i
  -0.1090 - 2.3664i
  -1.7820 +      0i

G_4_zeros = -3
-------------------
G_5_poles =

  -3.1365 +      0i
   0.5683 + 3.0400i
   0.5683 - 3.0400i

G_5_zeros = -4.5000
-------------------
```

Problem 2.A — Stability

$$G_1(s) : \text{stable} \tag{22}$$
$$G_2(s) : \text{stable} \tag{23}$$
$$G_3(s) : \text{unstable} \tag{24}$$
$$G_4(s) : \text{stable} \tag{25}$$
$$G_5(s) : \text{unstable} \tag{26}$$

Problem 2.B — Pole-zero sketches

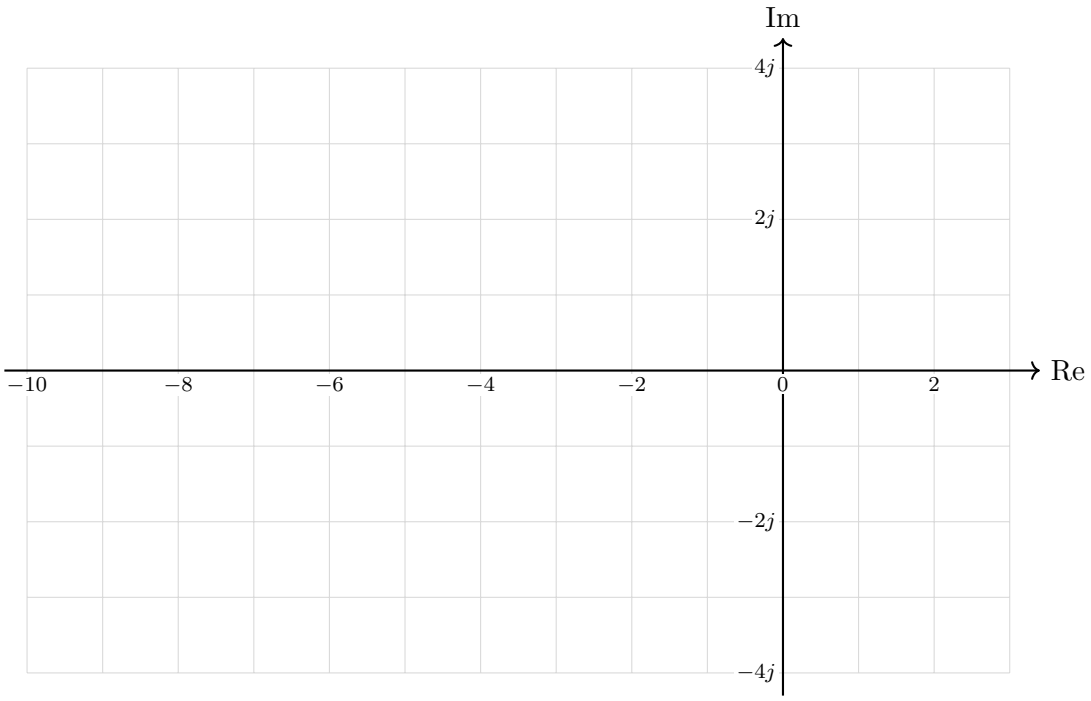


Figure 3: Pole-zero sketch of $G_1(s)$.

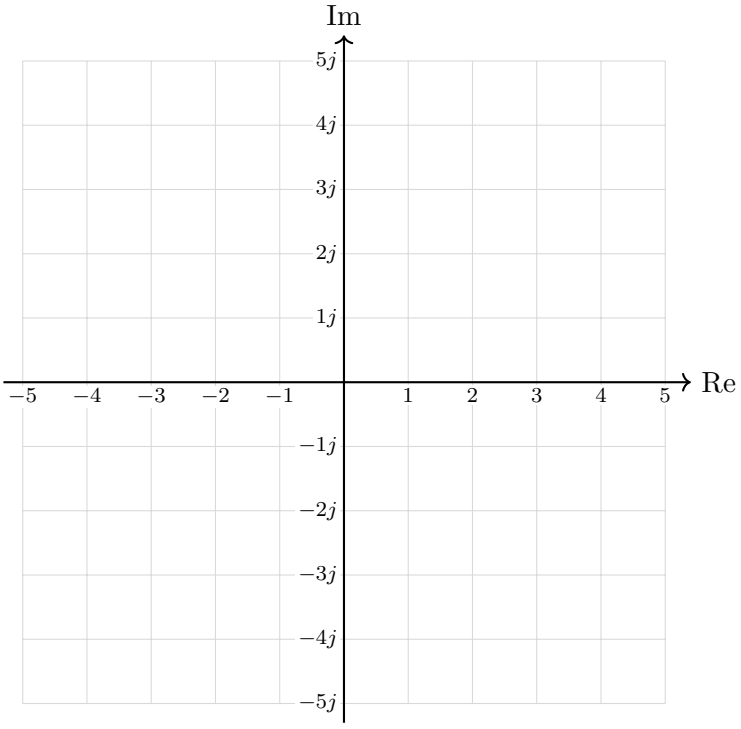


Figure 4: Pole-zero sketch of $G_2(s)$.

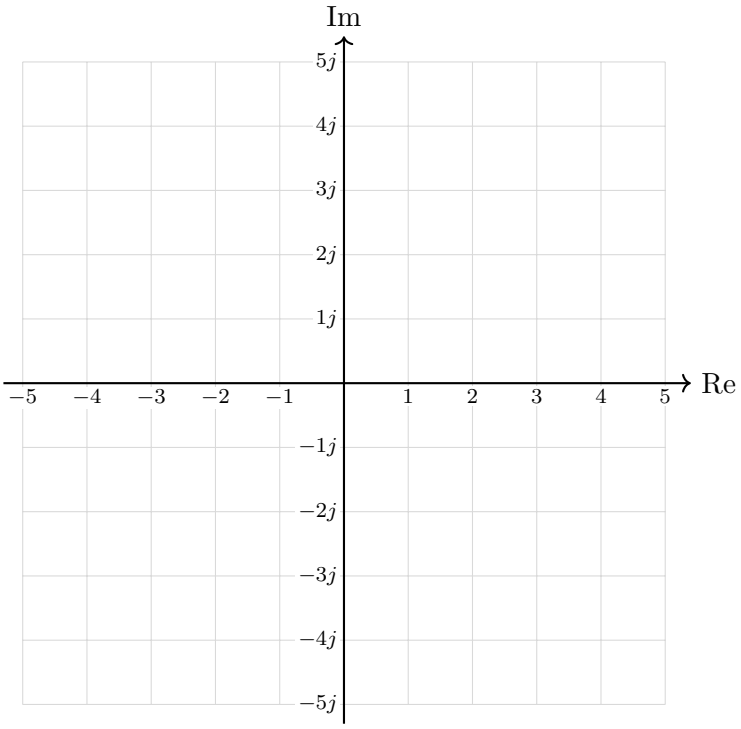


Figure 5: Pole-zero sketch of $G_3(s)$.

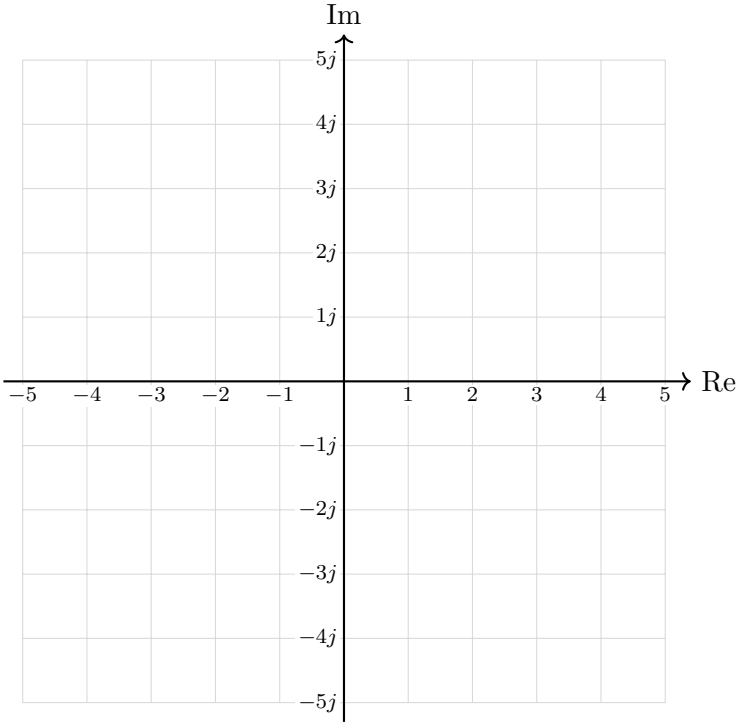


Figure 6: Pole-zero sketch of $G_4(s)$.

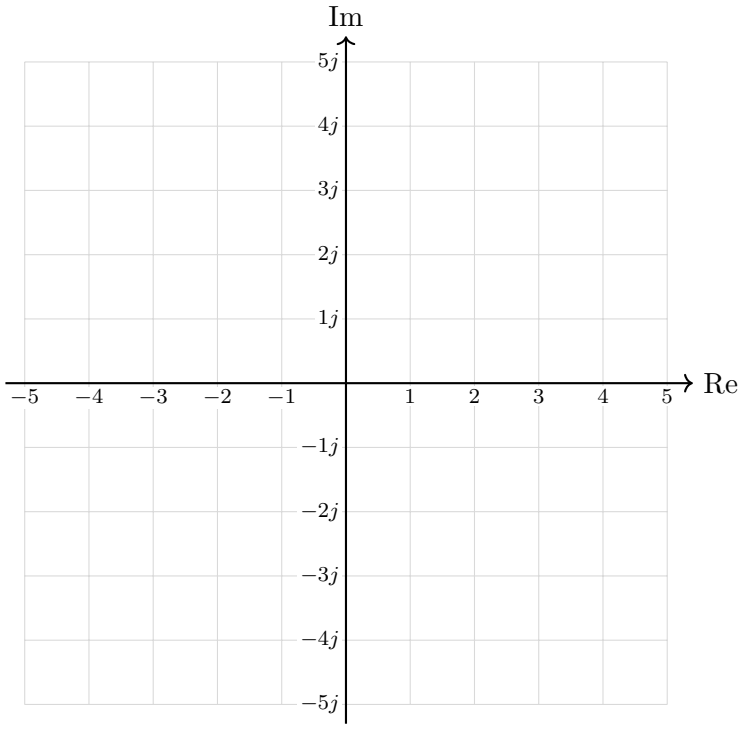


Figure 7: Pole-zero sketch of $G_5(s)$.

Problem 2.C — Damping factor

$$\alpha_2 = 1 \tag{27}$$
$$\omega_2 = \pm 3 \tag{28}$$
$$\omega_{2n} = \sqrt{\alpha_2^2 + \omega_2^2} \tag{29}$$
$$\omega_{2n} = \sqrt{10} \tag{30}$$
$$\rho_2 = \frac{1}{\sqrt{10}} \tag{31}$$

$$\alpha_4 = 0.109 \tag{32}$$
$$\omega_4 = \pm 2.3664 \tag{33}$$
$$\omega_{4n} = \sqrt{\alpha_4^2 + \omega_4^2} \tag{34}$$
$$\rho_4 = \frac{\alpha_4}{\sqrt{\alpha_4^2 + \omega_4^2}} \tag{35}$$

Code: `hw_05_2_C.m`

```
a_2 = 1;
w_2 = 3;

p_2 = a_2/sqrt(a_2^2 + w_2^2)

disp("-------")

a_4 = 0.109;
w_4 = 2.3664;

p_4 = a_4/sqrt(a_4^2 + w_4^2)
```

Output: `output/hw_05_2_C.txt`

```
p_2 = 0.3162
-------
p_4 = 0.046013
```

Problem 3.a — Methods of Partial Fraction Expansion

By plug & chug:

$$F(s) = \frac{1}{s(s+1)} \tag{36}$$
$$= \frac{C_1}{s} + \frac{C_2}{s+1} \tag{37}$$
$$= \frac{C_1(s+1) + C_2 s}{s(s+1)} \tag{38}$$
$$= \frac{(C_1 + C_2)s + C_1}{s(s+1)} \tag{39}$$

$$C_1 + C_2 = 0 \tag{40}$$
$$C_1 = 1 \tag{41}$$
$$C_2 = -1 \tag{42}$$

$$F(s) = \frac{1}{s} + \frac{-1}{s+1} \tag{43}$$
$$f(t) = 1^+(t) - e^{-t} \tag{44}$$

By cover-up:

$$C_1 = \big[F(s)s\big]_{s=0} = \big[\frac{1}{\cancel{s}(s+1)}\cancel{s}\big]_{s=0} \tag{45}$$
$$C_1 = 1 \tag{46}$$

$$C_2 = \big[F(s)(s+1)\big]_{s=-1} = \big[\frac{1}{s\cancel{(s+1)}}\cancel{(s+1)}\big]_{s=-1} \tag{47}$$
$$C_2 = -1 \tag{48}$$

Problem 3.c — Methods of Partial Fraction Expansion

$$F(s) = \frac{3s+2}{s^2+2s+10} \tag{49}$$

$$= \frac{3s+2}{(s+1-3i)(s+1+3i)} \tag{50}$$

$$= 3\,\frac{s+\frac{2}{3}}{(s+1-3i)(s+1+3i)} \tag{51}$$

$$K_{rlg} = 3 \tag{52}$$

Code: `hw_05_3_C.m`

```
if exist('OCTAVE_VERSION', 'builtin')
    pkg load control;
end

num = [3 2];
den = [1 2 10];

Zeros = roots(num)
Poles = roots(den)

F = tf(num, den);

figure;
pzmap(F);
legend off;
axis([-5 5 -5 5]);
axis square;
grid on;
title('Pole-zero map of F(s) = (3s+2)/(s^2+2s+10)');
```

Output: `output/hw_05_3_C.txt`

```
Zeros = -0.6667
Poles =

  -1 + 3i
  -1 - 3i
```

By cover-up:

$$F(s) = \frac{C_1}{s+1-3i} + \frac{C_2}{s+1+3i} \tag{53}$$

$$F(s)(s+1-3i) = C_1 + \frac{C_2(s+1-3i)}{s+1+3i} \tag{54}$$

$$C_1 = \Big[F(s)(s+1-3i)\Big]_{s=-1+3i} \tag{55}$$
$$= \Big[\frac{(3s+2)(s+1-3i)}{(s+1-3i)(s+1+3i)}\Big]_{s=-1+3i} \tag{56}$$
$$= \Big[\frac{(3s+2)}{6i}\Big]_{s=-1+3i} \tag{57}$$
$$= \frac{-3+9i+2}{6i} \tag{58}$$
$$= \frac{-1+9i}{6i} \cdot \frac{-i}{-i} \tag{59}$$
$$= \frac{9+i}{6} \tag{60}$$

$$C_2 = \Big[F(s)(s+1+3i)\Big]_{s=-1-3i} \tag{61}$$
$$= \Big[\frac{(3s+2)(s+1+3i)}{(s+1-3i)(s+1+3i)}\Big]_{s=-1-3i} \tag{62}$$
$$= \Big[\frac{(3s+2)}{-6i}\Big]_{s=-1-3i} \tag{63}$$
$$= \frac{-3-9i+2}{-6i} \tag{64}$$
$$= \frac{1+9i}{6i} \cdot \frac{-i}{-i} \tag{65}$$
$$= \frac{9-i}{6} \tag{66}$$

$$F(s) = \frac{9+i}{6(s+1-3i)} + \frac{9-i}{6(s+1+3i)} \tag{67}$$
$$f(t) = \frac{9+i}{6}e^{(-1+3i)t} + \frac{9-i}{6}e^{(-1-3i)t} \tag{68}$$

Using $e^{(-1\pm 3i)t} = e^{-t}(\cos 3t \pm i \sin 3t)$:

$$f(t) = \frac{e^{-t}}{6}\Big[(9+i)(\cos 3t + i\sin 3t) + (9-i)(\cos 3t - i\sin 3t)\Big] \tag{69}$$
$$= \frac{e^{-t}}{6}\Big[(9\cos 3t + 9i\sin 3t + i\cos 3t - \sin 3t) + (9\cos 3t - 9i\sin 3t - i\cos 3t - \sin 3t)\Big] \tag{70}$$
$$= \frac{e^{-t}}{6}\Big[18\cos 3t - 2\sin 3t\Big] \tag{71}$$
$$= e^{-t}\left(3\cos 3t - \frac{1}{3}\sin 3t\right)1^{+}(t) \tag{72}$$

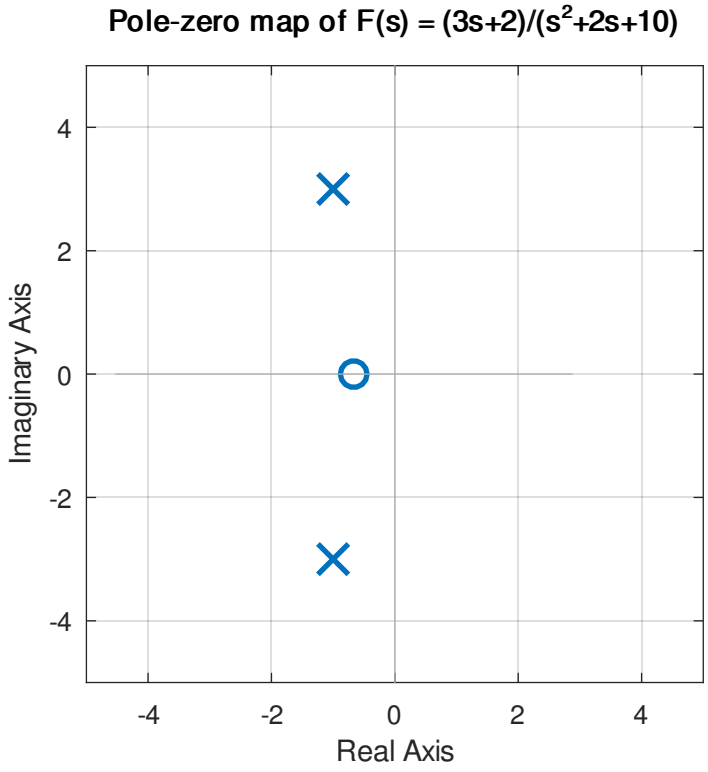


Figure 8: Pole-zero map of $F(s) = \frac{3s+2}{s^2+2s+10}$.

$$[z_1] = (-\frac{2}{3}, 0) \tag{73}$$
$$[p_1] = (-1, 3) \tag{74}$$
$$[p_2] = (-1, -3) \tag{75}$$

$$[v_{zp}] = [p_1 - z_1] = (-\frac{1}{3}, 3) \tag{76}$$
$$[v_{pp}] = [p_1 - p_2] = (0, 6) \tag{77}$$

$$|v_{zp}| = \sqrt{(\frac{1}{3})^2 + 3^2} \approx 3.018 \tag{78}$$
$$\theta_{zp} = 180^\circ - \arctan 9 \approx 96.34^\circ \tag{79}$$

$$|v_{pp}| = 6 \tag{80}$$
$$\theta_{pp} = 90^\circ \tag{81}$$

$$C_1 = K_{rlg}\frac{3.018\angle 96.34^\circ}{6\angle 90^\circ} \tag{82}$$
$$= 3(0.5031\angle 6.34^\circ) \tag{83}$$
$$\approx 1.509\angle 6.34^\circ \tag{84}$$
$$= 1.509(\cos 6.34^\circ + i \sin 6.34^\circ) \tag{85}$$
$$\approx 1.5 + 0.1667i \tag{86}$$

$$C_2 = \overline{C_1} \approx 1.509\angle -6.34^\circ \tag{87}$$
$$\approx 1.5 - 0.1667i \tag{88}$$